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Context-dependent dominance: Pollution sources vs. tree species in shaping leaf-deposited PM characteristics

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1. Abstract & Scope

Particulate matter (PM) characteristics are shaped by both biological and environmental variables. This study used neural-network source apportionment to determine whether the chemical characteristics of leaf-deposited PM are driven by regional pollution sources or species-specific traits. We sampled *Cinnamomum camphora* and *Sabina chinensis* leaves across Shanghai (industrial, traffic, residential, agricultural). In high-pollution traffic and industrial zones, pollution source characteristics completely overtook tree species traits, depositing uniform EC/OC (carbonaceous) chemical signatures on leaves. Species traits only dominated in low-pollution zones.

2. Methodology & Experimental Parameters

Foliar PM extraction. Analysis of carbonaceous fractions (EC/OC), water-soluble ions, and trace metals. Back-propagation neural network modeling for source apportionment.

3. Core Findings & Data Synthesis

Pollution source dominance overrides plant species traits in high-exposure zones. Disparate species accumulated uniform chemical profiles near highways. Species traits only dictate the chemical profile in clean zones.

4. Strategic Relevance to JU and Kolkata Planning

Confirms that in Kolkata's high-pollution zones, source characteristics will dominate chemical profiles, but species traits still dictate total mass capacity.

